# Ian Kim

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## EDUCATION

**Johns Hopkins University**, MSE in Computer Science — Baltimore, MD
*Graduation Date: December 2027*

**Williams College**, BA in Computer Science — Williamstown, MA
*GPA: 3.61/4.0* — *Graduation Date: June 2026*

- **Coursework:** Computer Security, Design of Experiments, Foundations of Artificial Intelligence, Algorithm Design and Analysis, Data Structures and Advanced Programming, Principles of Programming Languages, Computational Analysis of Big Data, Computer Organization, Discrete Mathematics, Statistical Modeling

## TECHNICAL SKILLS

**Programming Languages:** C, Python, SQL, R, C#, Java, JavaScript/TypeScript, HTML/CSS

**Tools/Frameworks:** React, Node.js, PostgreSQL, MongoDB, Git, Redis, Capacitor, Supabase, Unity Game Engine

## WORK EXPERIENCE

**WorldCare**, *Artificial Intelligence Research Intern* — *January 2026*
*Remote*
- Developed clinical data de-identification pipeline to process protected health information while ensuring HIPAA compliance; designed modular architecture using Python, Presidio, and Faker
- Engineered context-aware anonymization module and clinical filtering system to intelligently replace personally identifiable information while distinguishing medical terminology from protected health information
- Conducted research analyzing privacy and security risks of large language model deployment with protected health information; collaborated with two-person team to document PHI exposure vulnerabilities and present AI safety mitigation strategies to technical leadership

**Professor James Bern Lab Group**, *Research Assistant* — *June 2025 – Aug 2025*
*Williamstown, MA*
- Built a real-time interaction pipeline using Unity, Teensy microcontroller, and ODrive motor controller to simulate tactile object collisions with robotic haptic feedback
- Developed a C-based driver program to coordinate VR object tracking, collision detection, and motor control in real time
- Conducted controlled experiments measuring haptic feedback accuracy and motor response latency; collaborated with a four-person team to refine the mixed reality system

**Williams College**, *Teaching Assistant - Data Structures & Algorithms* — *September 2025 – June 2026*
*Williamstown, MA*
- Led weekly labs covering core data structures, algorithms, and computational complexity
- Guided students through debugging, algorithm optimization, and reasoning about performance tradeoffs
- Reviewed student code and provided structured feedback focused on correctness, clarity, and efficiency

## PROJECTS

**PokéNet**, *Personal Project* — *August 2026*
- Built a fine-grained image recognition system for 151 Pokémon species using deep metric learning, embedding images into a 128-D similarity space via a frozen OpenCLIP backbone with a trained projection head
- Fixed embedding collapse in triplet loss training by redesigning the mining strategy and loss function, unlocking stable convergence
- Hit 90% top-1 accuracy, an 18-point jump over the zero-shot CLIP baseline, using a k-means clustered reference gallery for nearest-neighbor lookup

**RedPrompt**, *Computer Security Final Project* — *May 2026*
- Engineered a local AI application environment using Ollama and Streamlit to simulate a realistic deployment target for prompt injection attack research
- Developed a formal testing framework to systematically evaluate prompt injection efficacy against a locally hosted DeepSeek model, measuring attack success rates across multiple injection strategies

## LEADERSHIP & COMMUNITY ENGAGEMENT

**Williams Varsity Men's Lacrosse**, *Player* — *September 2022 - June 2026*
**Berkshire Center for International Policy**, *Member* — *September 2025 - June 2026*